Development of a Simplified Ground-Based Radiometer for Lunar and Solar Thermal Observation

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Abstract—Ground-based microwave radiometry offers a low-cost method for characterizing planetary surface thermal emission without active illumination and can inform the design of spaceborne instruments such as NASA's Lunar Microwave Active-Passive Spectrometer (L-MAPS), an Artemis-era instrument being developed to characterize subsurface ice and thermal structure at the lunar South Pole. This project developed and validated a simplified ground-based radiometer system built around a parabolic dish antenna, a software-defined radio (SDR) receiver, and custom Python data acquisition software, with the goal of detecting thermal emission from the Sun and Moon. Receiver assembly and software development were completed, followed by a two-point calibration process anchored to published brightness temperatures for cold sky and the Sun. Solar observations yielded brightness temperatures of 9,265–10,752 K, within about 8% of the expected ~10,000 K, and detections were confirmed at a second frequency while disappearing outside the receiver's passband, indicating the signal originates from the sky rather than internal electronics. Lunar detection remains pending completion of a beam-dilution correction and antenna pointing calibration, identified as the primary next steps toward validating the system against the Moon's fainter thermal signal.

Index Terms—Radiometry, lunar surface, microwave remote sensing, brightness temperature, L-MAPS, calibration.

I. INTRODUCTION

Ground-based microwave radiometry provides a passive, low-cost means of characterizing the thermal and dielectric properties of planetary surfaces without requiring active illumination of the target. Because microwave emission originates from within the uppermost layers of a regolith rather than strictly at the surface, as is the case for thermal-infrared emission, microwave brightness temperature measurements carry information about subsurface structure that infrared instruments cannot directly access [2], [3]. This property makes ground-based and orbital microwave radiometry a valuable complement to visible and infrared lunar surface studies, and motivates its continued use in current lunar exploration efforts.

Historically, thermal characterization of the Moon began with infrared thermocouple measurements. Pettit and Nicholson [3] used a vacuum thermocouple at the 100-inch telescope at Mount Wilson to measure the Moon's subsolar-point temperature at approximately 407 K at full phase, falling to 358 K near quarter phase, and documented rapid surface cooling during a lunar eclipse consistent with a poorly conducting, low-thermal-inertia surface layer. Piddington and Minnett [2] extended this work into the microwave regime, measuring the Moon's disk-integrated thermal emission at a wavelength of 1.25 cm (24,000 Mc/s) across several lunar cycles. They found a smaller temperature amplitude (approximately ±40–52 K) and a phase lag of roughly 45° relative to the visible lunar phase, in contrast to the much larger amplitude and negligible lag seen in the infrared. They attributed this behavior to microwave emission originating below the immediate surface, within a partially transparent subsurface layer—some of the earliest evidence for the layered, low-thermal-inertia structure of the lunar regolith.

This subsurface sensitivity remains central to modern orbital microwave radiometry of the Moon. Siegler et al. [1] combined multi-frequency brightness temperature measurements from the Chang'E-2 microwave radiometer with thermal models constrained by the Lunar Reconnaissance Orbiter Diviner infrared radiometer to derive a frequency-dependent microwave loss tangent for the lunar regolith, relating microwave penetration depth to regolith composition, including titanium- and ilmenite-bearing terrain. Because different microwave frequencies probe different depths within the regolith, multi-frequency radiometry can be used to reconstruct a coarse subsurface thermal and compositional profile—a capability with direct application to characterizing subsurface volatiles near the lunar poles.

This capability underlies the motivation for the Lunar Microwave Active-Passive Spectrometer (L-MAPS), a NASA Artemis-era flight instrument under development at the University of Hawai‘i at Mānoa in collaboration with the Jet Propulsion Laboratory (JPL) and international partners. L-MAPS is intended to characterize subsurface ice and thermal structure at the lunar South Pole to depths exceeding 10 meters. The work described in this report was undertaken as a stepping stone toward that goal: the development and validation of a simplified, low-cost ground-based microwave radiometer, intended to test measurement and calibration techniques applicable to the passive microwave spectrometer component of L-MAPS using a parabolic dish antenna and commercially accessible RF hardware.

II. SYSTEM DESIGN AND METHODS

A. Literature Review and System Familiarization

Prior to hardware development, published lunar radio observation literature, including [1]–[3], and Dr. Matthew Siegler's work on lunar microwave emission, together with L-MAPS instrument team materials, were reviewed to document the expected signal chain of the ground-based system and to identify performance requirements ahead of dish integration. Four concepts from this review were applied directly to the system design and calibration procedure below: the radiometer equation, Friis noise cascade analysis, Dicke switching, and the Rayleigh–Jeans approximation.

For a total-power radiometer, the minimum detectable temperature change is given by the radiometer equation,

$$\Delta T_{min} = T_{sys} / \sqrt{B \cdot \tau} \tag{1}$$

where T_{sys} is the system noise temperature, B is the receiver bandwidth, and τ is the integration time. This relation guided the choice of sample rate and integration time used during data acquisition, described in Section II-B.

The receiver's overall noise contribution was estimated using the Friis noise cascade formula,

$$F_{total} = F1 + (F2-1)/G1 + (F3-1)/(G1G2) + \ldots \tag{2}$$

which shows that the noise contribution of each stage beyond the first is suppressed by the gain of the preceding stages, motivating the placement of a low-noise amplifier as the first active component in the signal chain.

In the Rayleigh–Jeans limit, appropriate for microwave frequencies and the physical temperatures of the Sun and Moon, the brightness temperature T_B of a source is related to its physical temperature T and emissivity ε by

$$T_B = \varepsilon \cdot T \tag{3}$$

which is the basis for converting calibrated radiometer output into a physically meaningful brightness temperature.

B. Receiver Assembly and Software Development

The receiver was assembled and bench-tested using a software-defined radio (SDR) connected to a Raspberry Pi. Data acquisition software was written in Python: the main program collects and records time-stamped power measurements, while a second script provides a real-time signal strength readout used while pointing the dish. Testing showed that operating the SDR at higher sample rates overflowed the Raspberry Pi's USB bus, so a sample rate of 2.048 Msps was adopted, logging 1,500 power samples over runs of 30 seconds or longer.

C. Calibration and Characterization

With the receiver assembled and logging, calibration and characterization were performed next. The receiver's gain was observed to drop gradually for the first several minutes after power-up, consistent with thermal warm-up of the receiver electronics. This was addressed by allowing the system to settle before recording, and by taking reference and target measurements back-to-back so that any residual drift would cancel out.

Cross-calibration of the dish against a known reference in an anechoic chamber was planned but could not be performed because the chamber was not available for use during the project period. A hot-and-cold reference calibration was used instead, with a nearby building serving as the warm reference and cold sky as the cold reference, to verify that the two reference points were stable and well-separated. The calibrated brightness temperature was computed from the two reference points using

$$T_B = T_{cold} + [(V-V_{cold})/(V_{hot}-V_{cold})] \cdot (T_{hot}-T_{cold}) \tag{4}$$

where V is the measured voltage for the target, and V_{cold} and V_{hot} are the voltages measured for the cold-sky and warm reference points, respectively. For solar calibration, the two reference points were anchored to published brightness temperatures of 20 K for cold sky and approximately 10,000 K for the Sun. Observations were cross-checked against published solar flux values, and measurements were matched at consistent air mass to minimize atmospheric effects. Because the angular size of the Sun is much smaller than the dish's beam width, a beam-dilution correction is required to recover an accurate brightness temperature; this correction was still in progress at the time of this report.

III. RESULTS

In an initial solar calibration session, the two reference points (a nearby building and cold sky) produced consistent, repeatable readings, confirming the reliability of the measurement scale. Using these references, raw solar readings were converted to brightness temperatures of 9,265–10,752 K across individual runs, within approximately 8% of the Sun's expected brightness temperature of ~10,000 K (Table I). Observations of blank sky produced values of +586 K and −532 K, near zero on the scale where the Sun reads 10,000 K, consistent with a properly functioning system (Fig. 1).

Target	Measured T_B (K)
Sun, run 1	9,265
Sun, run 2	10,752
Cold sky, run 1	+586
Cold sky, run 2	−532

TABLE I. Solar and cold-sky brightness temperature measurements from the initial calibration session.

[Fig. 1 — insert image here]

Fig. 1. Two-point calibration (top and middle) and the resulting brightness-temperature recovery for Sun and sky runs (bottom).

This result was confirmed at a second frequency (1420 MHz), which produced a similarly clear solar detection. Tuning to 1750 MHz, outside the receiver's normal operating range, produced only flat noise with no solar signal, indicating that the detected signal originates from the sky through the antenna rather than from internal electronics.

An attempt to reproduce the initial solar detection was unsuccessful; no measurable difference between Sun and sky pointing was observed, which was attributed to poor antenna positioning in a busy radio-frequency environment. To rule out the receiver gain setting as the cause, a sweep of gain values was performed, identifying a gain of 15.7 as the setting that reliably produced a clear solar detection without saturating the receiver (Fig. 2); this gain value has been used for all subsequent measurements.

[Fig. 2 — insert image here]

Fig. 2. Left: at the original gain, the receiver is saturated and the Sun (orange) is indistinguishable from cold sky (blue). Right: at reduced gain, the Sun is clearly detected.

Because the Moon's expected brightness temperature is only a few hundred kelvin, well within the ±600 K noise observed in the sky measurements, lunar detection has not yet been attempted with confidence; it requires completion of the beam-dilution correction and improved antenna pointing accuracy.

IV. DISCUSSION

The solar calibration results demonstrate that the ground-based radiometer system can reliably detect and quantify thermal emission from a strong, well-characterized source. The recovered brightness temperatures bracket the expected solar value within about 8%, a reasonable result for a system built from commercially accessible components. The persistence of the detection across two well-separated frequencies, and its disappearance outside the receiver's passband, provides strong evidence that the measured signal is astronomical in origin rather than an artifact of the receiver electronics.

The irreproducible result from the second observation session highlights the sensitivity of the system to antenna pointing and local radio-frequency interference, and underscores the importance of the gain optimization performed afterward. The resulting gain setting of 15.7 has since produced consistent results and will be used going forward.

The primary limitation preventing lunar detection at this stage is the mismatch between the dish's beam width and the angular size of both the Sun and Moon. Because the beam is much wider than either source, raw brightness temperature measurements represent a beam-averaged value rather than the true source temperature, an effect that is proportionally more significant for the Moon, whose expected signal (a few hundred kelvin, consistent with the amplitudes reported by [2] and [3]) is smaller than the ±600 K noise floor observed in blank-sky measurements. Completing the beam-dilution correction, together with a more precise antenna pointing calibration, is necessary before lunar brightness temperatures can be measured with confidence.

A. Future Work

Several steps remain to complete the project's lunar observation goals. First, the beam-dilution correction must be completed by properly characterizing the beam solid angle and the source solid angle, so that measured temperatures reflect the true source temperature rather than a beam-averaged value; all lunar measurements depend on this correction. Second, the dish's true pointing offset must be determined, since an offset-fed dish does not point where its face points; this will be resolved by using the Sun to locate the offset (peaking the signal and noting the difference) and then applying it when aiming at the Moon. Third, a minimum of three lunar observation sessions will be conducted across different lunar phases in late July through early August, using back-to-back reference and target runs to cancel warm-up drift, and tracking on- and off-source each time to confirm that the signal follows the Moon rather than the surrounding sky. Finally, the resulting brightness temperatures will be compared against published lunar datasets, including the phase-dependent amplitude reported by Piddington and Minnett [2], to check whether they vary with phase in the expected pattern, with any discrepancies investigated as potential systematic error sources.

V. CONCLUSION

At the midpoint of this project, the full radiometer signal chain has been assembled and bench-tested, and a working two-point calibration process anchored to published cold-sky and solar brightness temperatures has been established. Solar calibration converted raw readings into brightness temperatures of 9,265 K and 10,752 K across individual runs, bracketing the expected ~10,000 K within about 8%. This detection held up at a second frequency and disappeared entirely outside the receiver's passband, providing confidence that the system is detecting genuine solar thermal emission rather than an artifact of the equipment itself. Additional work remains before the system can be trusted to measure the Moon, a target with an expected brightness temperature only a few hundred kelvin above cold sky, far fainter than the Sun's ~10,000 K signal. Completing the beam-dilution correction and improving antenna pointing accuracy are the two primary remaining steps toward reliable lunar observation.

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